United International **University** (**UIU**)

Dept. of Computer Science and Engineering (CSE)

Final Term Assessment Year: **2025** Semester: **Summer**

Course: **CSE 3711** Title: **Computer Networks (Section – ALL)**

Marks: **40** Time: **2 Hour**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **Four (04)** questions. Answer **all of them**. All questions are of values indicated on the right-hand margin.

IMPORTANT NOTE: Candidates are required to answer the questions in sequential order (Q1, Q2, Q3, and Q4).

Q1.

a) Suppose, **Host A**, with **IP address 192.168.100.21** and **source port 5555**, sends an **HTTP request** to **google.com**, a web server that gives secured **HTTP service** using IP address **172.217.19.164** and port number **443**, and **Host B**, with **IP address 172.16.111.10** and **source port 8888**, sends an **SMTP request** to **gmail.com**, an email server that gives web-based **email service** using IP address **142.251.16.109** and port number **587**. **Describe** how **multiplexing** and **demultiplexing** occur at the Transport layer for the above scenario. **[4]**

b) For each of the **following functions**, tell whether it is implemented in **TCP, UDP, both,** or **none**: **[2]**
(i) data transfer reliability (ii) socket with port number (iii) security
(iv) connection management (v) multiplexing (vi) connectionless demultiplexing

c) **List at least two (2)** differences between **TCP** and **UDP header formats**. **[2]**

d) Suppose that the following four bytes are being sent via **UDP**: **[3]**

11101011 10100101 01011010 10010101

Now answer the following questions:

i. What will be the value of the **checksum** field in the UDP packet that is being sent?

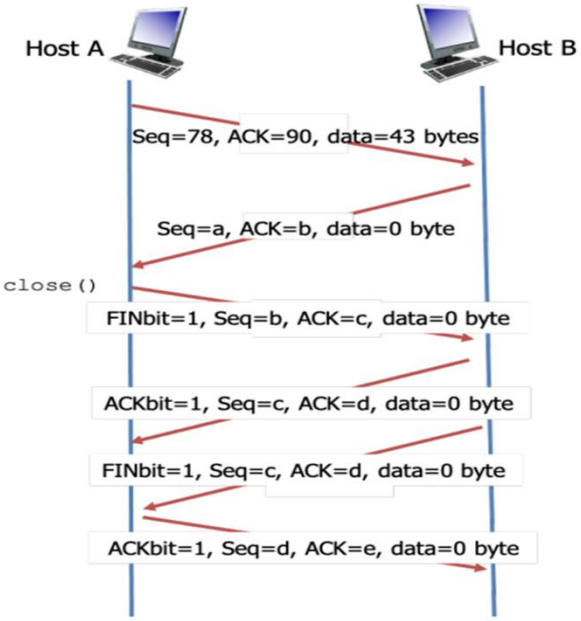
ii. Suppose the **3rd bit** of the **first byte** and **2nd bit** of the **third byte** of the data being sent is changed while the UDP packet is being transmitted. How will the receiver know that the data was corrupted? Explain your answer.

Q2.

a) Suppose, the **largest ACK** that a **TCP sender** has received is **A** and the most recently received **receiver window size** is **W**. If the **highest sequence number** it has sent so far is **H** then **how many more bytes** is it allowed to send? Give an **arithmetic expression** using variables **A, W,** and **H**. **[2]**

b) Briefly explain the **two techniques** for **retransmission** (timeout & 3-dup) discussed in class. What is the **advantage of fast retransmission** over the **timeout mechanism** in the original TCP? **Explain**. **[2]**

c) Following **diagram** shows a **sequence of packet exchange** in a **TCP connection** between **Host A** and **B**. **Host A** sends the **last data packet** and receives the **ACK packet** from **Host B**. Then, **Host A** sends a **FIN packet** to initiate closing of the connection. **Host B** sends a **FIN packet** as well. What are the values of **a, b, c, d, e**? **[5]**



Q3.

(a) Consider a following IPV4 datagram

ID	Total length	Fragment flag	Fragment offset
400	5500	0	0

arriving at a source and must be forwarded in fragments to transmit to the destination with an MTU of 1516 bytes. Illustrate all the fragmented datagrams by calculating the length, ID, fragment flag and offset of each datagram. Compute the additional overhead due to fragmentation.

[3+1]

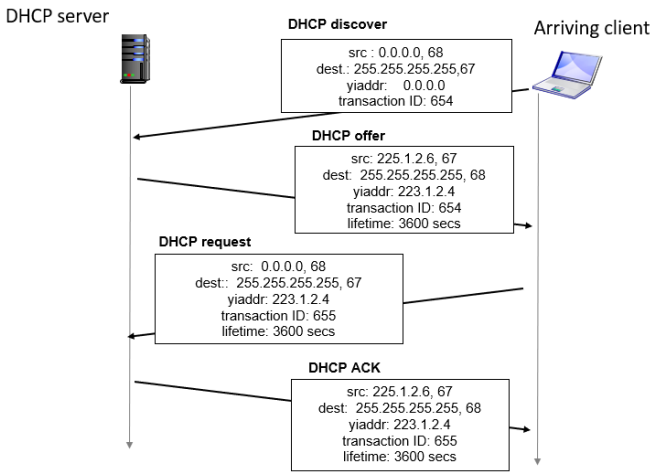


Figure 1: DHCP client-server scenario

(b) A DHCP client-server scenario is given in Figure 1. Answer the following queries:

i. What are the functions of DHCP?
ii. What is the purpose of using destination IP address 255.255.255.255 for DHCP discover message?
iii. What is the IP address of DHCP server?
iv. What is the final IP address assigned by DHCP server for the Arriving client?

[4]

(c) Draw a network scenario with the following scenario of NAT table:

[2]

NAT Table

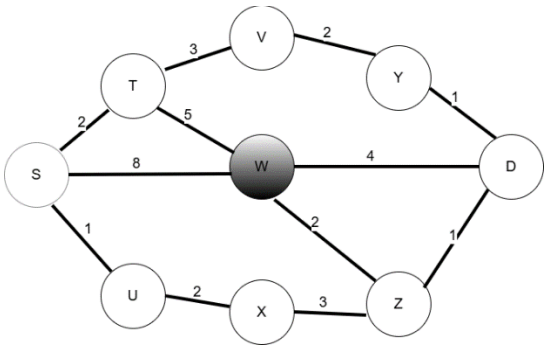
WAN side - IP, Port	LAN side- IP, Port
120.80.30.4, 3001	10.0.0.1, 241
120.80.30.4, 3002	10.0.0.2, 241
120.80.30.4, 3003	10.0.0.3, 241
120.80.30.4, 3004	10.0.0.1, 242

Now, show source and destination IP and port numbers before and after NAT for an email service from device with IP 10.0.0.1 and port 242 and destination IP 193.169.1.10 with port 25.

Q4.

(a) Given graph G = (V, E), where V is the set of routers and E is the set of links. Using Dijkstra's link-state routing algorithm, compute the least cost path from node U to all other nodes and show the resulting least-cost-path tree from U.

[5]



(b) The Data Link Layer is the second layer in the TCP/IP reference model. Answer the following questions:

i. What are the main services provided by this layer? Describe three of them briefly.

ii. Explain the process and identify the specific protocol utilized by a host on a local area network (LAN) to resolve the MAC address of a destination device, given only its IP address.

iii. Where is the link layer implemented?

[2+2+1]

END OF THE QUESTION